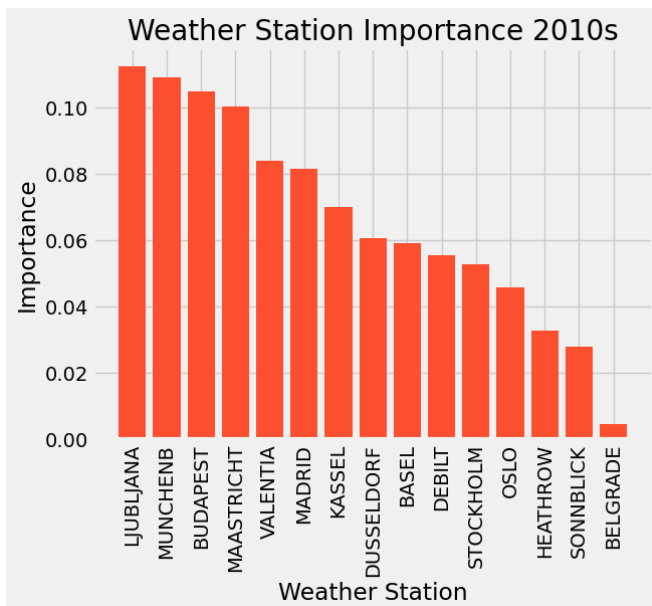
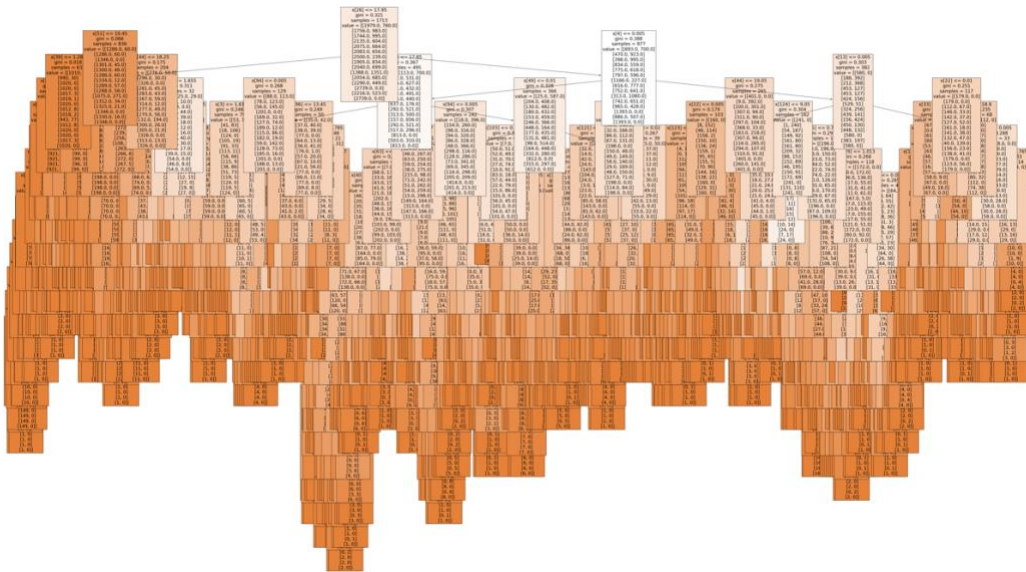


Exercise 2.3: Complex Machine Learning Models and Keras Part 2

Random Forest Model

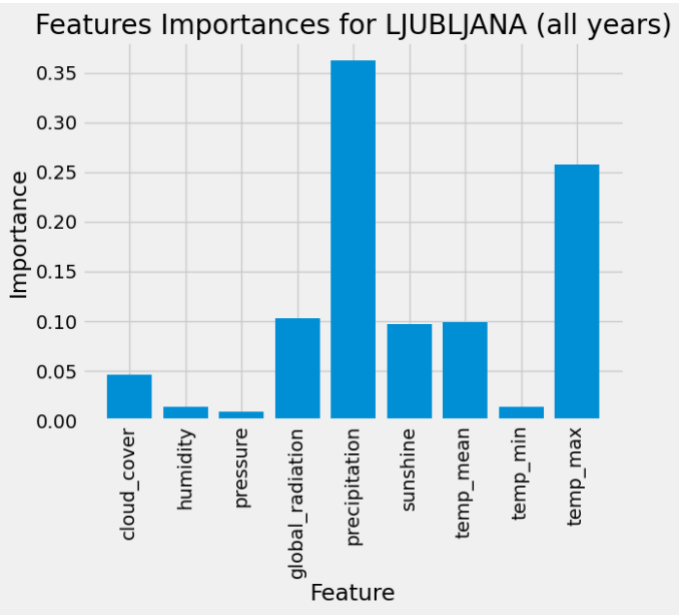
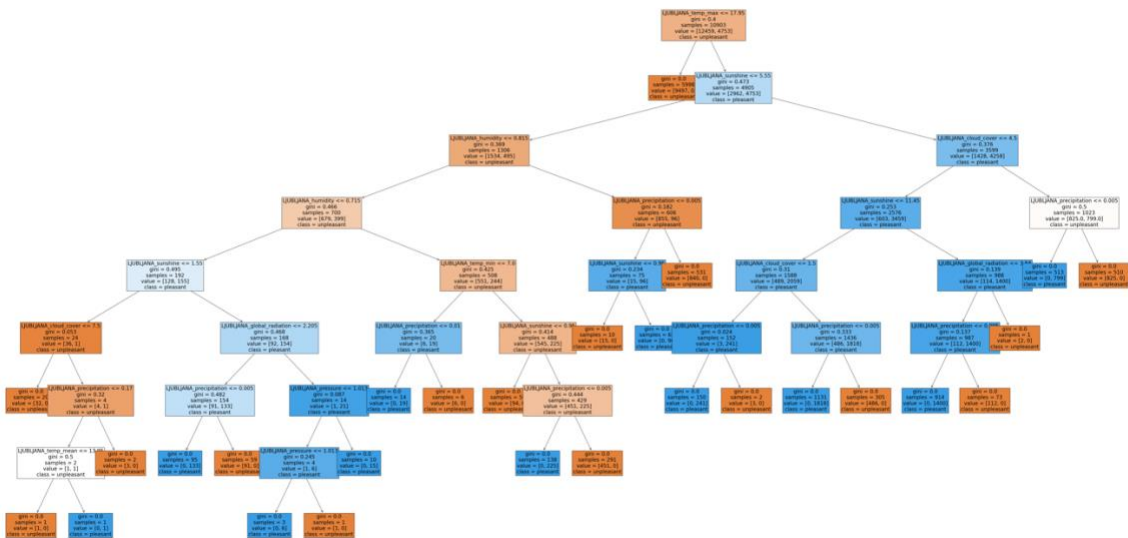


The dataset was limited to just the 2010s, and when training a Random Forest model with `n_estimators=100` and leaving `max_depth` to be determined automatically, the model achieved an accuracy of 58.7%. Unlike typical classification tasks where the `np.argmax()` method is used to convert one-hot encoded labels into class indices, this setup retained all 15 columns of the one-hot encoded target (`y` array). As a result, the decision trees had to handle a multi-output classification problem, which increased their complexity.

The 3 most important features resulted in: LJUBLJANA, MUNCHENB, BUDAPEST.

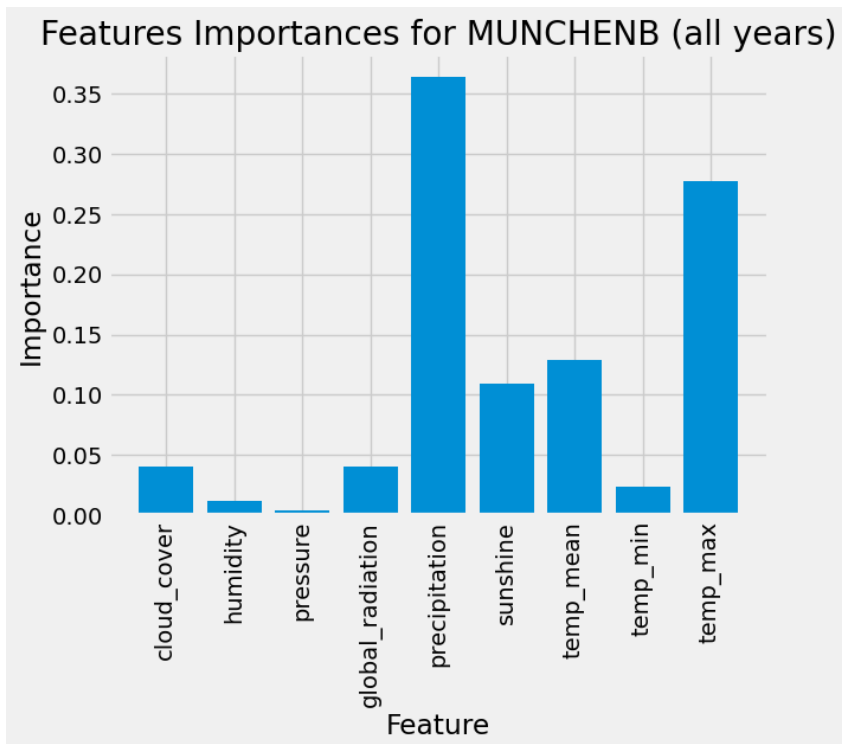
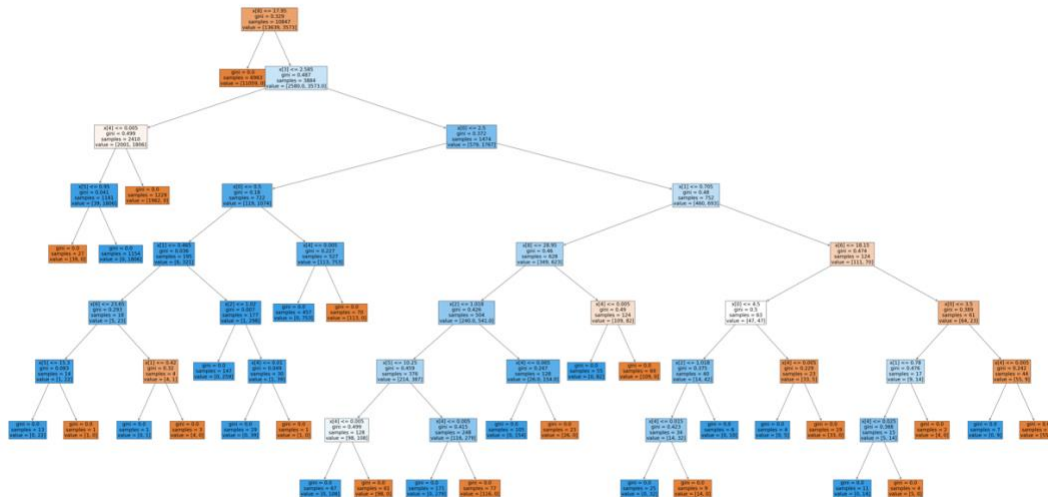
For each of the three stations, a separate Random Forest model was trained using the full range of available years in the observation dataset, rather than limiting the data to just a single decade. This approach allowed the models to learn from a broader temporal scope, potentially capturing more generalizable patterns and improving performance by leveraging the complete historical data.

1. LJUBLJANA



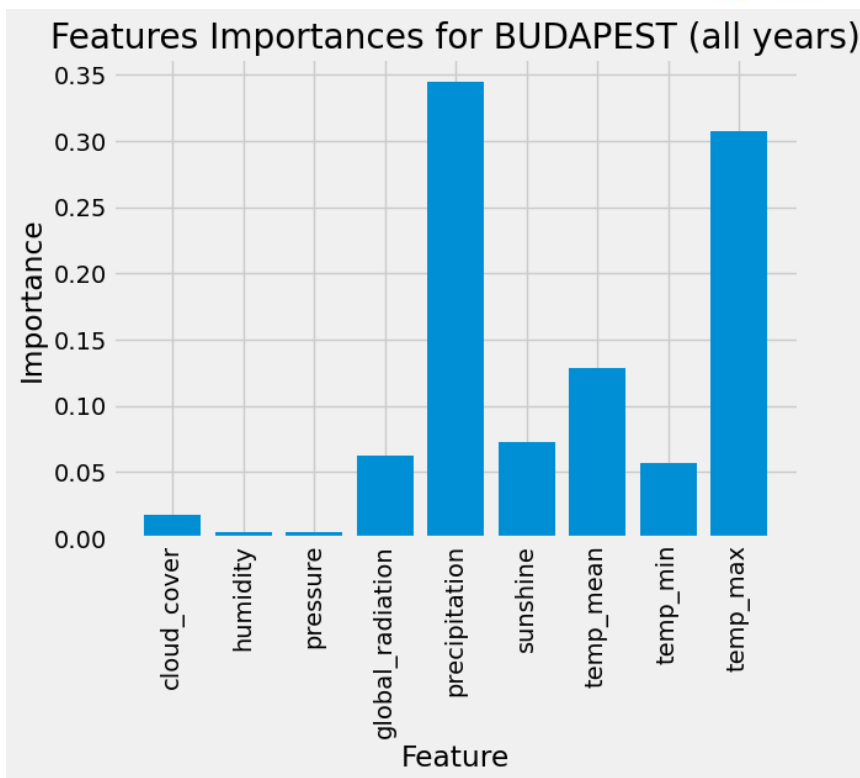
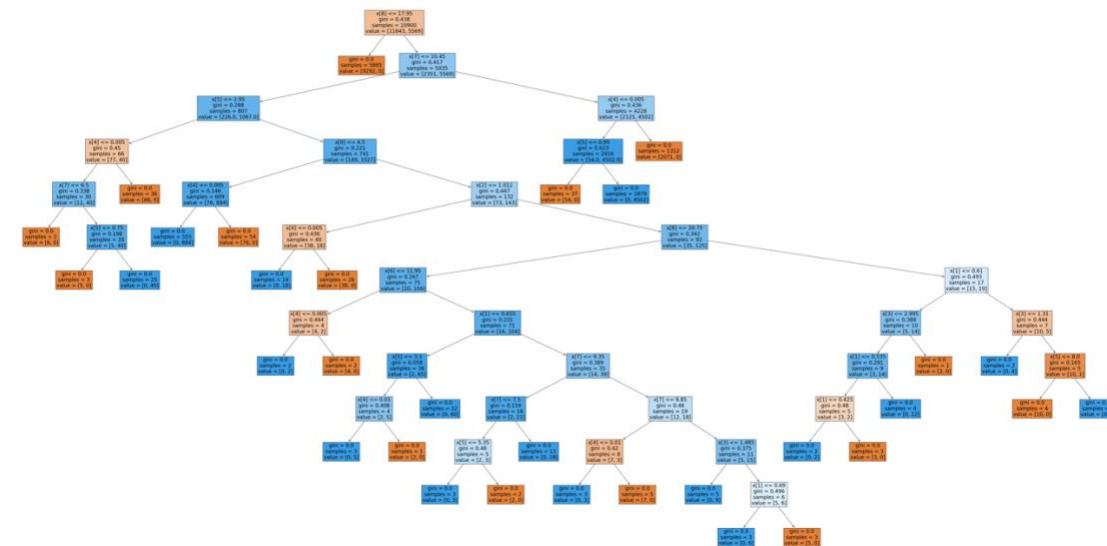
The most important factors were precipitation, followed by maximum temperature (temp_max) and global radiation. These features had the biggest influence on the model’s accuracy, while things like humidity and pressure mattered much less.

2. MUNCHENB



This chart shows which weather features were most important for predicting outcomes at the MUNCHENB station. The top three features were precipitation, maximum temperature (temp_max), and average temperature (temp_mean). These had the biggest impact on the model's decisions, while features like pressure and humidity mattered very little.

3. BUDAPEST



This chart shows which weather features were most important for predictions at the BUDAPEST station. The three most important ones were precipitation, maximum temperature (temp_max), and average temperature (temp_mean). These features played

the biggest role in helping the model make accurate predictions, while things like humidity and pressure had very little impact.

In **Conclusion**, looking at all three charts (LJUBLJANA, MUNCHENB, and BUDAPEST), we can see a clear pattern: precipitation and maximum temperature (temp_max) are the two most important weather features for all stations. Average temperature (temp_mean) also plays a strong role, though slightly less than the first two. Other features like humidity, pressure, and cloud cover are much less important for the model's predictions across all locations.